Meta-Collaborative Learning for Arbitrarily Scaled Hyperspectral Image Super-Resolution

Mingyang Zhang, *Member, IEEE*, Shuang Wu, Xiangyu Wang, Maoguo Gong, *Fellow, IEEE*, Fenlong Jiang, *Member, IEEE*, Yu Zhou, and Yue Wu, *Senior Member, IEEE*

***Abstract*— Deep learning-based methods for hyperspectral image super-resolution (SR) have achieved significant success in recent years. These methods typically consist of feature extraction module (FEM) and upsampling module. However, due to structural limitations of the upsampling module, most current methods focus on training separate models for different scale factors, which ignores the exploration of potential feature interdependence among different scale factors. In response to these challenges, we introduce a novel framework, called "meta-collaborative learning for arbitrarily scaled hyperspectral image super-resolution" (MCArb). Specifically, MCArb integrates a collaborative learning framework with a meta-learning-based 3-D upsampling module (3DMetaUM) and a scale-aware feature adaptation module (SAFAM). It enables training multiple SR tasks at different scale factors within a single network at the same time. This strategy is able not only to process arbitrary-scale-factor SR for hyperspectral images but also to harness the latent feature interdependence among different scales. In this study, we applied the MCArb framework to transform three deep learning-based hyperspectral image SR networks to MCArb methods, resulting in significant performance enhancements across five hyperspectral datasets. These improvements showcase the proposed MCArb framework's ability to enhance feature extraction efficiency and capitalize on latent interscale correlations. This code is available at https://github.com/ShuangWu-XDU/MCArb_HSI_SR.**

***Index Terms*— Collaborative learning, hyperspectral image, meta-learning, scale arbitrary, super-resolution (SR).**

I. Introduction

HYPERSPECTRAL images capture the reflectance information of various scenes across an extensive range of closely spaced spectral bands, spanning from tens to hundreds [1]. Compared to standard RGB or multispectral images (MSIs), hyperspectral images reveal subtle, intrinsic characteristics of objects that MSIs cannot detect. This unique capability has made hyperspectral imaging an essential tool in diverse fields, including object classification [2], medical diagnosis [3], detection processes [4], and geological exploration [5] within computer vision and remote sensing. However, due to sensor limitations, hyperspectral images often sacrifice spatial resolution for richer spectral information [6], resulting in reduced spatial detail compared to RGB images or MSIs. This tradeoff constrains the analysis and broader applications of hyperspectral images. Enhancing the spatial resolution of hyperspectral images through algorithmic approaches has become a critical focus in image processing, given the limitations of hardware sensors.

Image super-resolution (SR) technology restores detailed information and enhances data content in images, generating high-resolution (HR) imagery from low-resolution (LR) inputs via algorithms without modifying hardware sensors [7], [8], [9]. This approach addresses spatial resolution limitations of hyperspectral images. Hyperspectral SR methods are broadly classified into fusion-based methods [10], [11] (sometimes called hyperspectral pansharpening [12]) and single hyperspectral image SR (SHSR) [10], [11]. Fusion-based methods often outperform SHSR but require precisely aligned auxiliary data, which are impractical in many scenarios. In contrast, SHSR eliminates the need for auxiliary data, making it more applicable in real-world settings. However, traditional SHSR methods, including sparse representation [13], [14] and low-rank matrix models [15], [16], [17], [18], rely on fixed spatial and spectral priors, which limits their generalizability to complex, real-world conditions [19], [20]

Recently, several deep learning-based SHSR algorithms have achieved impressive results [21]. Especially, with the advancement of deep learning, end-to-end processing strategies are increasingly becoming popular, categorized into group-based methods, like [22], and direct end-to-end methods, like [23]. In addition, the use of 2-D or 3-D convolutions [23], transformers [20], [24], and modules based on attention mechanisms [25] in SHSR tasks has been explored. In terms of training loss design, researchers have also proposed various approaches, including spectral loss, total variation (TV) loss, adversarial loss [25], and perceptual loss [26], to enhance the performance and visual outcomes of SHSR.

Deep learning-based SHSR models typically consist of two key modules: a feature extraction module (FEM) and an upsampling module. While significant advancements have

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Mingyang Zhang, Shuang Wu, Xiangyu Wang, and Yu Zhou are with the School of Electronic Engineering, Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an 710071, China (e-mail: omegazhangmy@gmail.com; shuangwu@stu.xidian.edu.cn; wangxiangyu@stu.xidian.edu.cn; zhouyu@xidian.edu.cn).

Maoguo Gong is with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an 710071, China, and also with the College of Mathematics Science, Inner Mongolia Normal University, Hohhot 010028, China (e-mail: gong@ieee.org).

Fenlong Jiang and Yue Wu are with the School of Computer Science and Technology and the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an 710071, China (e-mail: jiangfenlong@outlook.com; ywu@xidian.edu.cn).

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been made in improving FEM, the upsampling module has emerged as a critical bottleneck. Conventional upsampling methods, such as interpolation-based techniques [19], transposed convolution [27], and pixel shuffle [28], [29], are restricted to fixed scale factors, forcing SHSR models to train separate networks for each scale. This design overlooks interscale feature interdependence, limiting model scalability and efficiency. To address these limitations, this study introduces a collaborative learning framework that optimizes multiple-scale SR tasks simultaneously during training, enabling a single network to handle arbitrary-scale SR through a meta-learning-based upsampling structure.

Collaborative learning, an emerging paradigm in deep learning, enhances performance by enabling multiple models or tasks to share knowledge and collaboratively optimize objectives. By facilitating information exchange, collaborative learning improves adaptability and efficiency, particularly in multitask scenarios. In the context of low-level vision tasks, such as image enhancement and reconstruction, this approach has demonstrated its effectiveness in leveraging shared features to boost performance across related tasks [30], [31], [32]. Inspired by these successes, this study extends collaborative learning to SHSR to address the limitations of conventional single-scale SR training frameworks. Collaborative learning provides an effective solution by enabling simultaneous optimization across multiple-scale SR tasks within a unified framework. This not only reduces redundancy but also helps discover latent interscale dependencies critical for improving reconstruction quality, especially at large scale factors. By adopting a collaborative learning framework, this study aims to overcome these limitations and unlock the full potential of multiple-scale feature sharing.

Meta-learning, often termed as "learning to learn," is increasingly notable in deep learning due to its adaptability [33]. Research in low-level vision tasks demonstrates the flexibility and efficacy of meta-learning frameworks, particularly in image enhancement and reconstruction [34], [35], [36]. Researchers such as Hu et al. [35], Wang et al. [37], and Fu et al. [38] have employed meta-learning-based methods in the field of RGB images SR. These methods focus on learning convolutional kernel weights tailored for different scale factors, achieving SR for arbitrary scale factors with a single network. Inspired by these works, we explore the introduction of meta-learning-based methods into SHSR. The motivation lies in meta-learning's ability to enable a single network to perform SR across multiple scale factors by dynamically predicting scale-specific parameters. Furthermore, integrating meta-learning into a collaborative learning framework allows simultaneous optimization of multiple-scale SR tasks, thereby fulfilling the intended purpose of collaborative learning.

Taking all the above issues into account, we propose a novel meta-collaborative learning framework (MCArb) to address the limitations of existing SHSR methods. MCArb enables arbitrary-scale SR within a single network while enhancing performance across multiple scales. Specifically, MCArb uses a collaborative learning framework to allow multiple SR tasks (corresponding to different scale factors) to be trained jointly, enabling effective feature sharing and uncovering interscale dependencies. At the core of MCArb is the meta-learning-based 3-D upsampling module (3DMetaUM), which dynamically predicts convolutional kernel weights for arbitrary-scale SR, replacing the static upsampling modules in baseline networks. To further enhance the performance, a 3-D spatial pyramid pooling module (3DSPPM) is integrated, facilitating feature exchange across task branches. In addition, a scale-aware feature adaptation module (SAFAM) is designed to address feature shifts observed during multiple-scale SR training, aligning feature representations dynamically across scales. Finally, an uncertainty loss strategy ensures balanced optimization across tasks. Together, these innovations allow MCArb to significantly improve SHSR performance, particularly for large scale factors like $8\times$. The main contributions of this work are summarized as follows.

1) *A Novel MCArb Framework:* By combining collaborative learning and meta-learning, this framework addresses the limitations of existing SHSR models, enabling arbitrary-scale SR and leveraging latent interscale correlations.
2) *Development of 3DMetaUM:* A flexible upsampling module that predicts scale-specific kernel weights dynamically, allowing a single network to handle arbitrary-scale SR tasks.
3) *Integration of 3DSPPM and SAFAM:* These modules ensure efficient intertask feature exchange and correct feature shifts during multiple-scale SR training, significantly improving SR performance for large scale factors.

The rest of this article is organized as follows. Section II briefly reviews recent SHSR background and introduces our motivation. Section III describes the designed framework in detail. We analyze and discuss the experiments in Section IV. Finally, Section V summarizes this work.

II. Background and Motivation

Recent advancements in SHSR have introduced sophisticated algorithms to address challenges in improving spatial resolution while preserving spectral accuracy. Methods leveraging symmetrical feature propagation strategies [39], dual-stage approaches [40], and RGB-induced feature modulation techniques [41] have demonstrated promising results by enhancing feature extraction and reconstruction fidelity. These methods focus on optimizing feature propagation and modulation, enabling SHSR models to better handle complex spatial–spectral correlations.

A. Collaborative Learning for SHSR

Collaborative learning has emerged as an effective approach to overcome the limitations of single-task optimization by enabling simultaneous optimization across related tasks [42]. Unlike traditional frameworks that train independent models for individual tasks, collaborative learning leverages shared representations to improve efficiency and adaptability in multitask scenarios. This paradigm has been successfully applied in various fields, including low-level vision tasks such as image enhancement and reconstruction [43], where it facilitates feature sharing across tasks to boost performance.

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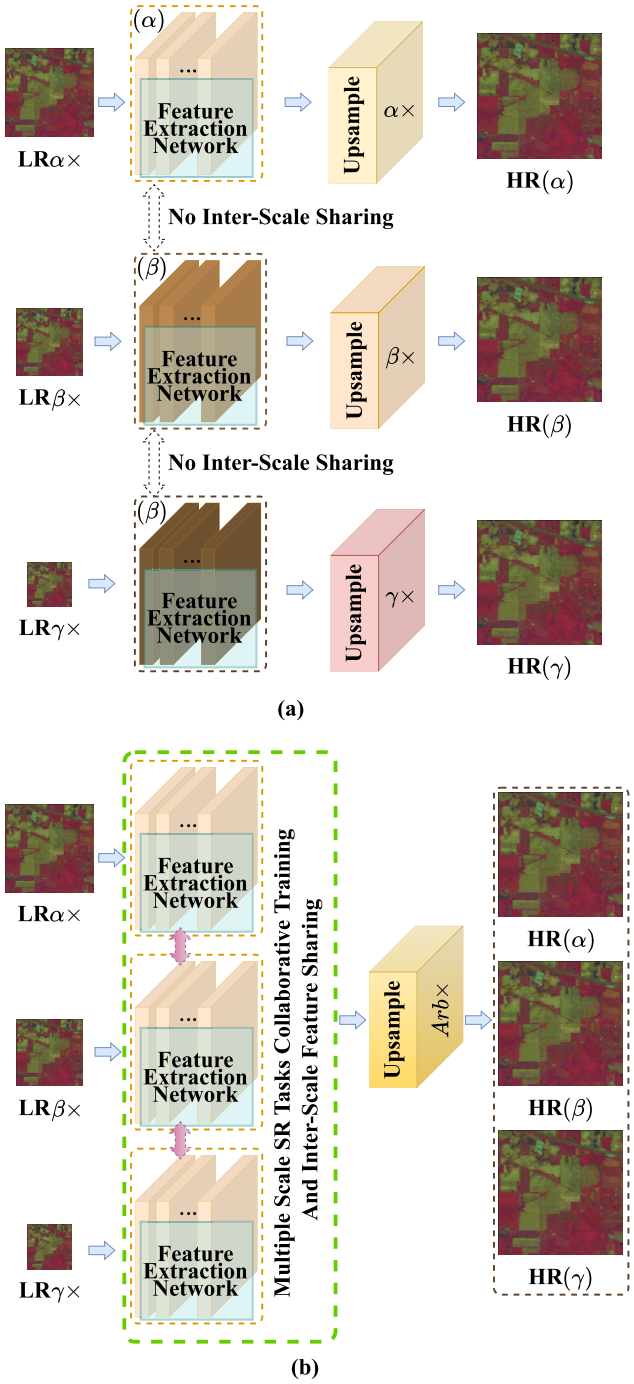


Fig. 1. Limitations of existing SHSR methods and the proposed framework. (a) Existing SHSR methods: isolated single-scale models. (b) Proposed collaborative framework.

As illustrated in Fig. 1(a), most existing SHSR methods rely on separate models trained independently for each scale factor. These models are designed to optimize performance for fixed-scale tasks, which restricts their ability to leverage latent interscale feature correlations. Furthermore, without intertask feature sharing, these models fail to exploit shared information across scale factors, limiting their flexibility and scalability in applications.

By introducing collaborative learning into SHSR, it becomes possible to optimize multiple-scale SR tasks within a unified framework, leveraging shared feature representations to improve reconstruction quality across diverse scales. As depicted in Fig. 1(b), the collaborative learning framework enables simultaneous training of SR tasks for different scale factors, promoting interscale feature sharing and addressing the rigidity of fixed-scale designs. This unified framework not only reduces redundancy but also ensures more efficient utilization of multiscale information, enhancing reconstruction quality in SHSR.

B. Meta-Learning for Upsampling in SHSR

Upsampling strategies play a crucial role in SHSR, as they directly determine the spatial resolution and reconstruction quality of the output images [44]. Commonly used methods include interpolation-based techniques, transposed convolution, and pixel shuffle. Interpolation-based techniques, such as bicubic interpolation or hybrid methods that combine interpolation and convolution layers, are computationally efficient but often oversmooth high-frequency details. Transposed convolution adaptively learns upsampling patterns through convolutional filters [45], but it is prone to introducing checkerboard artifacts, which can degrade image quality. Pixel shuffle rearranges feature maps to enhance spatial resolution and has been widely adopted in SHSR methods such as SSPSR [22] and HSRNet [46]. While effective, all these methods share a common limitation; they are inherently designed for fixed-scale operations. This fixed-scale constraint prevents the integration of multiple-scale SR within a single network. Overcoming this limitation requires a more adaptive approach to upsampling.

Inspired by advancements in RGB image SR, this study introduces a meta-learning-based upsampling module tailored to SHSR. This module is a key component of the collaborative learning framework depicted in Fig. 1(b), enabling dynamic adaptation to arbitrary scale factors. Traditional SHSR frameworks struggle to incorporate interscale sharing due to the rigidity of their upsampling modules, which are bound to fixed-scale operations. In contrast, our meta-learning-based module allows a single module to adapt to multiple-scale SR. The integration of the meta-learning-based upsampling module is pivotal for realizing the collaborative learning framework, as it breaks the reliance on fixed-scale designs and enables the simultaneous optimization of multiple-scale SR tasks. By addressing the limitations of existing upsampling methods, our framework is capable of leveraging shared information across scales while dynamically adapting to diverse scale factors, ultimately enhancing reconstruction quality in SHSR.

III. Proposed Method

This section presents the proposed MCArb framework for arbitrary-scale hyperspectral image SR. It is structured into five components: the overall framework, the 3DMetaUM module for flexible upsampling, the SAFAM module for scale-aware adaptation, the 3DSPPM module for feature alignment, and the loss function for multitask optimization. Each component is detailed as follows.

We can denote the original HR hyperspectral image as $I_{\text{HR}} \in \mathbb{R}^{H\times W\times B}$, where $H \times W$ is the spatial resolution of

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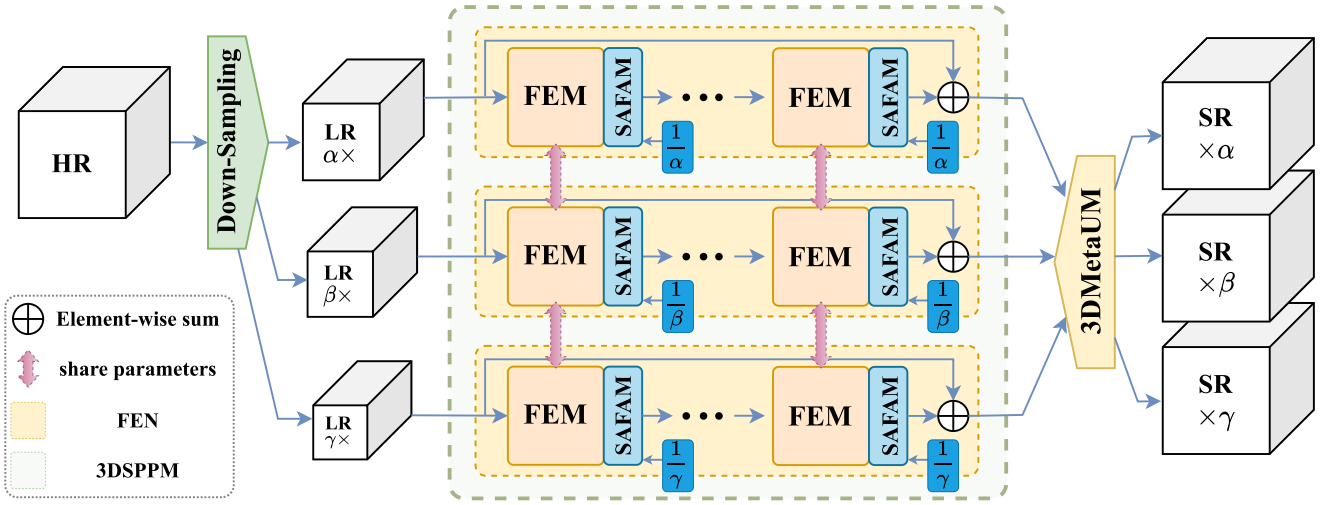


Fig. 2. Overall architecture of the proposed MCArb framework, including 3DMetaUM, SAFAM, and 3DSPPM. In our framework, three task branches are trained collaboratively, with the FEN sharing parameters across these branches. To correct the feature shift among different scale branches, the SAFAM is integrated into the baseline FEN. In addition, the 3DSPPM is employed to facilitate the flow of information among different branches. The framework utilizes a single 3DMetaUM to achieve arbitrarily scaled upsampling.

I_{HR} and B is the number of spectral bands. We downsample the I_{HR} by scale factors r to obtain the LR hyperspectral image $I_{\mathrm{LR}} \in \mathbb{R}^{H/r \times W/r \times B}$. The hyperspectral image SR network is designed to restore I_{LR} to $I_{\mathrm{SR}} \in \mathbb{R}^{H \times W \times B}$, aiming for I_{SR} to closely approximate I_{HR}. The relationship among these three images can be denoted as follows:

$$I_{\mathrm{LR}} = D(I_{\mathrm{HR}}, r) \tag{1}$$

$$I_{\mathrm{SR}} = R(I_{\mathrm{LR}}, r) \tag{2}$$

where $D(\cdot)$ represents the degradation model and $R(\cdot)$ denotes the SR reconstruction model.

As we discussed earlier, current deep learning-based SHSR models usually consist of feature extraction and upsampling modules. The process of converting LR images into HR images is illustrated as follows:

$$I_{\mathrm{HR}} = H_{\mathrm{UM}}(H_{\mathrm{FEN}}(I_{\mathrm{LR}})) \tag{3}$$

where H_{UM} and H_{FEN} represent the functions of upsampling modules and FEMs, respectively.

The structure of FEM generally does not limit the input and output sizes. For example, two LR images, I_{LR1} and I_{LR2} of sizes $H/r_1 \times W/r_1 \times B$ and $H/r_2 \times W/r_2 \times B$, respectively, are degraded from an HR image I_{HR} of size $H \times W \times B$ by different scale factors r_1 and r_2. After the FEM, they result in feature maps, of sizes $A(H/r_1 \times W/r_1) \times C$ and $A(H/r_2 \times W/r_2) \times C$, denoted as Fmap_1 and Fmap_2, respectively. A represents the size transformation function due to convolution, and C is the number of feature map channels. Most current methods adjust settings with stride and padding so that A generally keeps image sizes as follows:

$$A(x) = x \tag{4}$$

where x represents spatial size like $H/r_1 \times W/r_1 \times B$ and $H/r_2 \times W/r_2 \times B$. However, the structure of typical upsampling methods is only designed for specific scale factors separately, whether transposed convolution or pixel shuffle, like designing U_1 and U_2 to restore Fmap_1 and Fmap_2 to I_{HR}, respectively. The size transformation function of U_1 and U_2 is as follows:

$$U_1(x) = r_1 \times x \tag{5}$$

$$U_2(x) = r_2 \times x. \tag{6}$$

We can see that the typical upsampling methods confine current SHSR models to designing and independently learning models for specific scale factors, limiting arbitrarily scaled SR and the exploration among different scale factors in a single model.

A. Collaborative Learning Framework

To leverage the exploration of potential feature interdependence among different scale factors and to account for the consistent representation in the spatial and spectral contents reconstructed from different scale factors, as shown in Fig. 2, we propose a novel collaborative learning framework MCArb, aimed at reforming the architecture and training process of typical deep SHSR models.

During the training phase, an HR hyperspectral image I_{HR} is first downsampled by scale factors α, β, and γ (where $\alpha < \beta < \gamma$), resulting in $I_{\mathrm{LR}\alpha}$, $I_{\mathrm{LR}\beta}$, and $I_{\mathrm{LR}\gamma}$. These three LR images are then processed through feature extraction networks (FENs) in their respective branches, producing feature maps Fmap_α, Fmap_β, and Fmap_γ. FEN is derived from the baseline deep SHSR models. Our MCArb framework aims at developing a single network for multiple-scale-factor SR tasks. The corresponding FEMs in the FENs across different task branches share parameters. We observe a feature shift phenomenon when FENs process different LR images downsampled from the same HR image. Therefore, each FEM is closely followed by an SAFAM which is scale-aware, to correct this shift caused by different downsampling scale factors. The feature shift phenomenon will be specifically illustrated and discussed in Section III-C.

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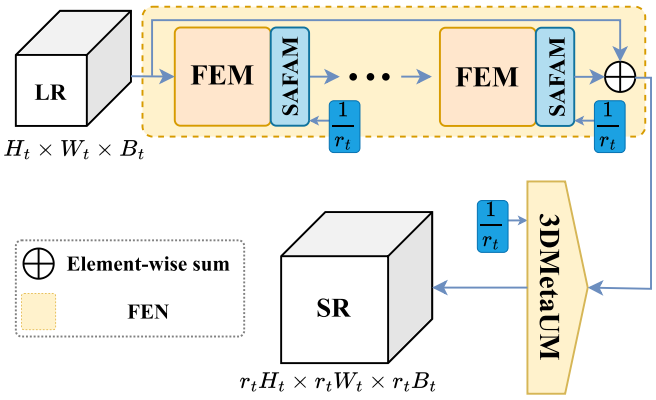


Fig. 3. Testing phase of our framework.

The entire feature extraction process can be represented as

$$\text{Fmap}_\kappa^1 = H_{\text{SAFAM}}^1\left(H_{\text{FEM}}^1(I_{\text{LR}\kappa}, \kappa)\right), \quad \kappa = \alpha, \beta, \gamma \tag{7}$$

$$\text{Fmap}_\kappa^N = G\left(H_{\text{SAFAM}}^N\left(H_{\text{FEM}}^N\left(\text{Fmap}_\kappa^{N-1}\right), \kappa\right)\right), \quad \kappa = \alpha, \beta, \gamma. \tag{8}$$

Fmap_κ^1 signifies the feature map processed by the first FEM and SAFAM, while Fmap_κ^N denotes the feature map processed by the Nth FEM and SAFAM. $G(\cdot)$ represents intermodule operations, such as residual and dense connections from the baseline model. The functions of FEM and SAFAM are, respectively, denoted as $H_{\text{FEM}}(\cdot)$ and $H_{\text{SAFAM}}(\cdot)$.

Subsequently, inspired from meta-learning research [35], [37], [38], we design a 3DMetaUM to enable upsampling arbitrary scale factors within a single module. Through our 3DMetaUM upsampling, three SR images of the same size as HR are obtained from different branches, $I_{\text{SR}\alpha}$, $I_{\text{SR}\beta}$, and $I_{\text{SR}\gamma}$ as follows:

$$I_{\text{SR}\kappa} = H_{\text{MetaUM}}\left(\text{Fmap}_\kappa, \kappa\right), \quad \kappa = \alpha, \beta, \gamma \tag{9}$$

where $H_{\text{MetaUM}}(\cdot)$ denotes the function of the 3DMetaUM.

We acknowledge that in SHSR, tasks in larger scale factors are more research-oriented and present more challenges than those in smaller ones. Therefore, in the MCArb training process, the $\alpha\times$ task branch is regarded as an auxiliary task, while the other two are considered primary tasks. During this process, input to the $\alpha\times$ task branch introduces additional detail information to the FEN. This is leveraged to enhance the learning effectiveness of other branches through the weight sharing strategy and the feature information flow mechanism provided by the 3DSPPM.

In the testing phase, as shown in Fig. 3, given a test LR image I_{LRtest} of size $H_t \times W_t \times B_t$ and the required magnification factor r_t, we obtain the reconstructed SR image I_{LRtest} of size $r_tH_t \times r_tW_t \times r_tB_t$ through the MCArb model.

B. Meta-Based 3DMetaUM

Convolutional upsampling module reconstructs the Fmap into I_{SR}. Thus, each pixel $p_{\text{SR}}(i, j)$ in I_{SR} is determined by the corresponding pixel $p_F(i', j')$ in Fmap and the weight of

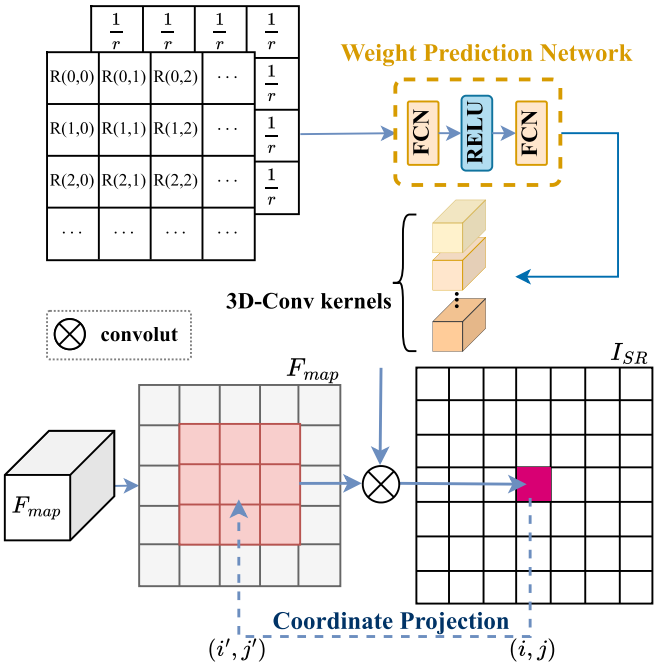


Fig. 4. Structure of the proposed 3DMetaUM.

corresponding convolution kernel, denoted as $W(i, j)$. We formulate this process as

$$p_{\text{SR}}(i, j) = H_{\text{SR}}\left(p_F\left(i', j'\right), W(i, j)\right) \tag{10}$$

where $H_{\text{SR}}(\cdot)$ represents the convolutional process.

This suggests that the upsampling module is designed to train weights to learn mapping between corresponding position pixels in Fmap and I_{SR}. Typically, conventional upsampling module learns such mappings at a fixed scale factor, and thus, the weights of module are also fixed with specific scale factor after training. This inherent limitation restricts them to SR hyperspectral images only at a fixed scale factor. Meta-learning, often referred to as "learning to learn," could offer a suitable solution to this limitation. Therefore, by employing the 3DMetaUM in Fig. 4, we can use one trained meta-learning-based network to dynamically predict the weights of 3-D convolutional kernels for different scale factors. Notably, in contrast to the 2-D convolution-based meta-learning upsampling used in natural images [35], 3-D convolution is more apt for processing feature maps from hyperspectral images. As a very important module in realizing our MCArb framework, 3DMetaUM enables a single network for arbitrarily scaled upsampling.

Specifically, we first need to find the corresponding projection of each pixel $p_{\text{SR}}(i, j)$ in I_{SR} on the pixel $p_F(i', j')$ in the Fmap as follows:

$$\left(i', j'\right) = \left(\left\lfloor \frac{i}{r} \right\rfloor, \left\lfloor \frac{j}{r} \right\rfloor\right) \tag{11}$$

where $\lfloor\rfloor$ denotes the floor function.

As illustrated in Fig. 5, the coordinate projection relationship can be more clearly observed, taking a scale factor of $2\times$ as an example. Each $p_{\text{SR}}(i, j)$ corresponds to a unique $p_F(i', j')$, and the final determination of the pixel $p_{\text{SR}}(i, j)$

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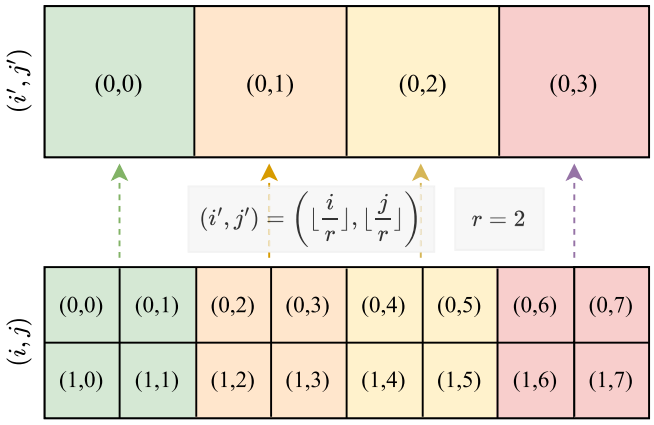


Fig. 5. Details of the coordinate projection relationship at 2×.

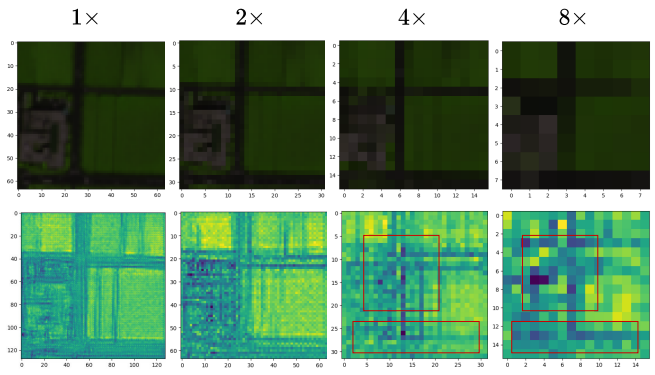


Fig. 6. First column of four images represents the HR image along with its LR counterparts, obtained through downsampling at scales of 2×, 4×, and 8×. The second column depicts feature maps extracted from the first column's images using a thoroughly trained SSPSR network.

also requires knowledge of the associated convolution kernel weight $W(i, j)$, which is predicted by our 3DMetaUM as follows:

$$W(i, j) = H_{\text{WP}}(R(i, j), r) \tag{12}$$

where $H_{\text{WP}}(\cdot)$ denotes the function of weight prediction network and $R(i, j)$ is the related distance derived from the coordinate projection, which can be expressed as follows:

$$R(i, j) = \left(\frac{i}{r} - \left\lfloor \frac{i}{r} \right\rfloor, \frac{j}{r} - \left\lfloor \frac{j}{r} \right\rfloor\right). \tag{13}$$

According to this function, when training SR tasks for scale factors with common divisors, the related distance matrices for different scale factors can overlap. For example, the related distance matrix at a 2× scale is a subset of those for 4× and 8× tasks. To avoid the collapse in weight prediction caused by this overlap, we also incorporate the scale factor into input of the weight prediction network as shown in Fig. 4. The final input of 3DMetaUM is as follows:

$$\text{Input}(i, j) = \left(\frac{i}{r} - \left\lfloor \frac{i}{r} \right\rfloor, \frac{j}{r} - \left\lfloor \frac{j}{r} \right\rfloor, \frac{1}{r}\right). \tag{14}$$

C. Scale-Aware Feature Adaptation Module

As mentioned earlier, we observed a feature shift phenomenon during feature exaction. Specifically, the Fmaps obtained by the same FEN should ideally exhibit consistency no matter the input I_{LR}s are downsampled from the same I_{HR} at any scale factors, that is, irrespective of spatial resolution of input I_{LR}, the high-level representations in the feature maps should maintain consistency. Nevertheless, as depicted in Fig. 6, when employing a single FEN from the fully trained SSPSR [22] for feature extraction on one patch of the Chikusei dataset, the corresponding Fmaps for different resolutions exhibit a pronounced shift, failing to maintain consistency. This inconsistency is particularly highlighted in the red box. We believe that this issue arises because the feature extraction logic of a single network struggles to adapt to the degradation logic of multiple scale factors. The larger the scale factor, the more information is lost, and the more feature shift occurs, as shown in the results at 8×.

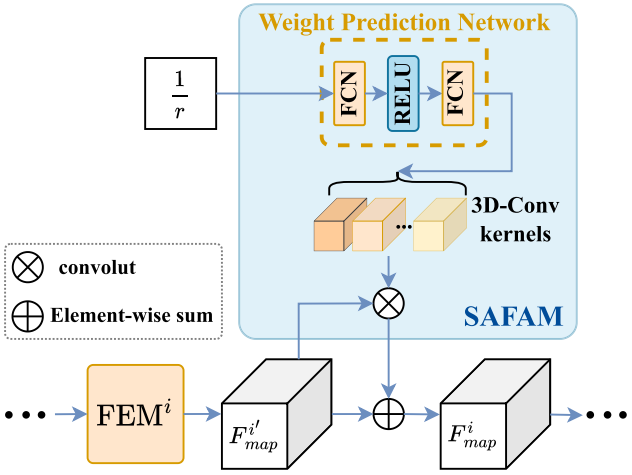


Fig. 7. Structure of SAFAM.

While the FEN is identical across branches in our framework, a module is needed to be aware of different scales and correct the corresponding shift across these branches. Similar to 3DMetaUM, we propose another meta-learning-based module, the SAFAM, which is integrated after each FEM in the FEN. As depicted in Fig. 7, this module accepts scale factor scalar as input and predicts 3-D convolutional kernel weights through a fully connected network. These kernels then convolve with the Fmaps obtained from each FEM. SAFAM is designed to produce distinct weights for different scale factor branches, thereby correcting the feature shifts caused by each FEM from the baseline's FENs. The SAFAM functions as a mapping mechanism between its inputs—scale factors—and its outputs, the convolutional kernels. In other words, SAFAM is used as a scale-aware module to perceive different scales and generate different convolutional kernels. While FEMs in different branches are the same because of sharing parameters, SAFAMs operate independently and are scale factor-aware. Section III-D will report how the 3DSPPM supervises SAFAM in learning feature shift information.

D. 3-D Spatial Pyramid Pooling Module

In our MCArb framework, we try to minimize the distance between feature maps of those branches to supervise SAFAM

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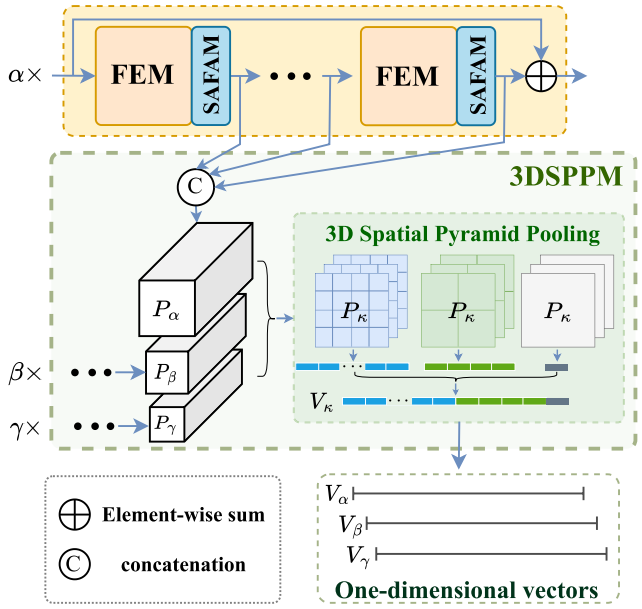


Fig. 8. Detail of the 3DSPPM.

keeping feature consistency and facilitate information flow across multiple branches. This allows smaller scale factor tasks branch to guide the learning process of larger ones. However, a direct comparison of similarity between feature maps from different branches is challenging due to their varying sizes. Spatial pyramid pooling, a prevalent technique in CNNs, adeptly addresses this issue [47], [48], [49], [50]. It can process input images of arbitrary sizes and produce output vectors of a fixed size. Therefore, in this article, we employ the 3-D spatial pyramid pooling method to effectively bridge this gap.

Specifically, as illustrated in Fig. 8, in the FEN, feature maps Fmap_κ^i (where i ranges from 1 to N, and $\kappa = \alpha, \beta, \gamma$) obtained from each layer of the FEMs across all branches are concatenated to form P_κ of different sizes. Through the 3DSPPM, we obtain three 1-D vectors V_κ of the same size, which can represent those feature maps Fmap_κ^i. Then, we minimize the distance among those V_κ. The corresponding formulas are as follows:

$$P_\kappa = \text{Cat}\left(\text{Fmap}_\kappa^i, \ldots, \text{Fmap}_\kappa^N\right), \quad \kappa = \alpha, \beta, \gamma \tag{15}$$

$$V_\kappa = H_{\text{SPPM}}(P_\kappa), \quad \kappa = \alpha, \beta, \gamma \tag{16}$$

where $\text{Cat}(\cdot)$ and $H_{\text{SPPM}}(\cdot)$, respectively, denote the function of concatenation and the 3DSPPM.

E. Loss Function

To enhance the SR effectiveness and to achieve a closer resemblance between the super-resolved image I_{SR} and the HR image I_{HR}, we investigate several loss functions in this article. Primarily, both $L1$ and $L2$ losses are considered at the pixel-wise level, commonly employed for direct point-to-point error computation in image tasks [51]. Owing to the inherent properties of the mean squared error (mse) function, $L2$ loss tends to produce overly smooth error results. In the context of SR tasks, such smoothing can inadvertently lead the network to generate excessively smooth outcomes, which is undesirable. Numerous studies have demonstrated that the mean absolute error (MAE) characteristics of $L1$ loss are more suited for SR tasks. Consequently, in this article, we select the $L1$ loss as our primary pixel-wise loss function, which is articulated as follows:

$$I_{\text{SR}\kappa} = H_{\text{MCArb}}(I_{\text{LR}\kappa}, \kappa), \quad \kappa = \alpha, \beta, \gamma \tag{17}$$

$$\mathcal{L}_1(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\left\|I_{\text{HR}}^n - I_{\text{SR}\kappa}^n\right\|_1, \quad \kappa = \alpha, \beta, \gamma \tag{18}$$

where Θ, N, and n represent the parameter of our network, the number of images, and the nth image, respectively, and H_{MCArb} denotes the total function of our MCArb network.

To suppress noise in the reconstructed image, drawing from established research experience [25], [52], our approach incorporates the TV loss, a metric evaluating the differences between adjacent pixels, into the pixel-wise loss function as follows:

$$\begin{aligned}\mathcal{L}_{\text{TV}}(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{C\times(H-1)\times(W-1)} \\ \times\sum_{i=1}^{C}\sum_{j=1}^{H-1}\sum_{k=1}^{W-1}\left(\left(I_{\text{SR}\kappa}^n(i,j+1,k) - I_{\text{SR}\kappa}^n(i,j,k)\right)^2\right. \\ \left.+\left(I_{\text{SR}\kappa}^n(i,j,k+1) - I_{\text{SR}\kappa}^n(i,j,k)\right)^2\right), \quad \kappa=\alpha,\beta,\gamma\end{aligned} \tag{19}$$

$$\mathcal{L}_{\text{pix}}(\Theta) = \mathcal{L}_1 + 1\times 10^{-6}\mathcal{L}_{\text{TV}}. \tag{20}$$

Although pixel-wise loss effectively constrains the network's training for SR tasks, it tends to overlook the spectral characteristics inherent in hyperspectral images, potentially leading to spectral distortion. To mitigate this phenomenon and preserve spectral details, we also introduce the spectral angle mapper (SAM) loss, articulated as follows:

$$\mathcal{L}_{\text{spe}}(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{\pi}\arccos\left(\frac{I_{\text{HR}}^n\cdot I_{\text{SR}\kappa}^n}{\left\|I_{\text{HR}}^n\right\|_2\cdot\left\|I_{\text{SR}\kappa}^n\right\|_2}\right) \quad \kappa=\alpha,\beta,\gamma. \tag{21}$$

Up to this point, we can obtain the overall loss for each task branch by weighting $\mathcal{L}_{\text{pix}}$ and $\mathcal{L}_{\text{spe}}$ at a certain ratio, formulated as follows:

$$\mathcal{L}_\kappa(\Theta) = \mathcal{L}_{\text{pix}} + 0.1\mathcal{L}_{\text{spe}}, \quad \kappa = \alpha, \beta, \gamma. \tag{22}$$

During the training phase of our framework, in addition to the sum of losses $\mathcal{L}_\kappa$ for each branch, it is also imperative to consider the objective function $\mathcal{L}_{\text{tri}}$, generated by 3DSSPM, which measures the differences between feature maps across branches as

$$\mathcal{L}_{\text{tri}}(\Theta) = \min_{V_\alpha, V_\beta, V_\gamma}\left(\|V_\alpha - V_\beta\|_2 + \|V_\alpha - V_\gamma\|_2 + \|V_\beta - V_\gamma\|_2\right). \tag{23}$$

The overall loss function for the entire framework is thus formulated as follows:

$$\mathcal{L}_{\text{total}}(\Theta) = \lambda_1\mathcal{L}_\alpha + \lambda_2\mathcal{L}_\beta + \lambda_3\mathcal{L}_\gamma + \lambda_4\mathcal{L}_{\text{tri}}. \tag{24}$$

We observe that the loss function $\mathcal{L}_{\text{total}}$ bears resemblance to a multitask learning scenario, where the magnitude and

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gradients of different losses can vary significantly. Assigning static values to λ_i, for $i = 1, 2, 3, 4$, could potentially deteriorate the results due to poor weighting. The model's learning efficacy is highly sensitive to these λ values, making it challenging to simultaneously achieve optimal performance across multiple task branches. Manual adjustment of these weights is time-consuming and often ineffective. Therefore, we model the uncertainty within individual tasks to guide the adjustment of these weights [53] as follows:

$$\mathcal{L}_{\text{total}}(\Theta, \lambda_i) = \frac{1}{2\lambda_1^2}\mathcal{L}_\alpha + \frac{1}{2\lambda_2^2}\mathcal{L}_\beta + \frac{1}{2\lambda_3^2}\mathcal{L}_\gamma + \frac{1}{2\lambda_4^2}\mathcal{L}_{\text{tri}} + \sum_{i=1}^{4}\log\lambda_i. \quad (25)$$

In essence, by dynamically adjusting the weights based on the observed uncertainty, our model avoids the pitfall of overfitting to more complex tasks while neglecting simpler ones. This method ensures a more equitable distribution of learning resources.

IV. Experiments and Results

This section evaluates the performance of the proposed MCArb framework through comprehensive experiments conducted on five datasets. We compared state-of-the-art methods with the MCArb methods that use the former as baselines in terms of quantitative metrics and visual quality. Furthermore, ablation studies evaluate the individual contributions of each module within the proposed method.

A. Implementation Details

In the proposed MCArb framework, input HR hyperspectral images are downsampled into three samples and collaboratively trained through three branches, setting as $\alpha\times = 2\times$, $\beta\times = 4\times$, and $\gamma\times = 8\times$. The $\alpha\times$ branch, serving as an auxiliary task, is designed to enhance the performance of $\beta\times$ and $\gamma\times$ networks. Consequently, only an equivalent of 25% of the samples is used for training in the $\alpha\times$ branch (the impact of the training sample proportion in the $\alpha\times$ branch on network performance will be discussed in subsequent experiments). In the 3DMetaUM and SAFAM, meta-learning-based weight prediction networks generate $5 \times 3 \times 3$ dimensional 3-D convolutional kernels. During 3-D spatial pyramid pooling in 3DSSPM, the height size of the convolutional kernel is also set to 5. In the uncertainty loss function, initial weights are set as $\lambda_i = 1$, for $i = 1, 2, 3, 4$. Regarding training settings, we employ the Adam optimizer [54] to train the network for 100 epochs. The initial learning rate is set to $1.5e^{-4}$ and decays tenfold after 65 epochs. The mini-batch size is set to 8. Our framework is implemented in Pytorch[1] on NVIDIA RTX 3090 GPUs.

B. Datasets and Experimental Setup

We analyze and evaluate our framework on three widely recognized public hyperspectral image datasets: Pavia Centre [55],[2] Chikusei [56],[3] and Houston 2018 [57].[4] To further validate the robustness and applicability of our framework across diverse conditions and spectral characteristics, we also selected two widely used real-world MSI datasets: CAVE [58][5] and Harvard [59].[6] This article is dedicated to transforming existing SHSR models using our framework. We select state-of-the-art models—MCNet [23], SSPSR [22], and ERCSR [27]—to provide the FEN. Specifically, the MCNet [23] model refers to a mixed 2-D/3-D convolutional network, employing a combination of 2-D and 3-D convolutional networks, sharing spatial information to better reconstruct spatial details. The learning spatial–spectral prior for SR (SSPSR) [22] model first upsamples the grouped subimages and then performs a secondary upsampling on the complete image by fusing the interpolated subimages. The ERCSR [27] means exploring the relationship between 2-D and 3-D convolution. This model alternates between 2-D and 3-D units to mitigate structural redundancy, thereby enhancing the model's learning capability in the spatial domain through the sharing of spatial information. These SHSR models encompass both end-to-end (MCNet [23] and ERCSR [27]) and group-based (SSPSR [22]) feature extraction strategies, including transposed convolution (MCNet [23] and ERCSR [27]) and subpixel convolution (SSPSR [22]) upsampling methods.

The transformed networks are referred to as MCArb-SSPSR, MCArb-MCNet, and MCArb-ERCSR. These MCArb methods will be utilized to demonstrate our framework's adaptability to different feature extraction strategies and its superiority over those upsampling methods. Notably, SSPSR [22] employs progressive upsampling, setting the upsampling layer at multiple stages of the network, including feature exaction stage. To control variables in our experiments, we introduce an additional variant of SSPSR [22] that utilizes backend upsampling, referred to as SSPSR-b. For comparative algorithms, we meticulously adjust their hyperparameters to optimize performance on the aforementioned datasets at scale factors of $4\times$ and $8\times$. Our fully trained MCArb method requires only a single network model to accomplish SR tasks at different scale factors, and its performance is compared against the original networks and traditional nondeep learning techniques, including wavelet transform, bilinear, and bicubic interpolation.

To evaluate the SR effectiveness, we employ six commonly used metrics: peak signal-to-noise ratio (PSNR), structure similarity (SSIM), cross correlation (CC), root-mean-squared error (RMSE), SAM, and normalized global error (ERGAS). The PSNR, SSIM, and RMSE, typically utilized for assessing natural image restoration quality, are calculated on single-channel images and are reported with mean values computed across all spectral channels. The remaining three metrics—CC, SAM, and ERGAS—are extensively applied in

[1][Online]. Available: https://pytorch.org

[2][Online]. Available: https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_scene

[3][Online]. Available: https://www.sal.t.u-tokyo.ac.jp/hyperdata/

[4][Online]. Available: https://hyperspectral.ee.uh.edu/?page_id=1075

[5][Online]. Available: https://cave.cs.columbia.edu/projects/categories/project?cid=Computational+Imaging&pid=Generalized+Assorted+Pixel+Camera

[6][Online]. Available: http://vision.seas.harvard.edu/hyperspec/

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TABLE I
QUANTITATIVE PERFORMANCE OVER EIGHT TEST IMAGES WITH DIFFERENT SCALE FACTORS ON THE PAVIA CENTRE DATASET. BOLD REPRESENTS THE BEST RESULTS

Scale	Method	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	Bilinear	27.2646	0.7241	0.8358	11.3694	6.3908	7.3401
	Bicubic	28.0022	0.7822	0.8553	10.4073	6.1378	6.7618
	Wavelet	27.0440	0.7407	0.8151	11.6768	6.5023	7.5232
	MCNet	29.2401	0.8478	0.8905	8.9893	5.7786	5.8864
	MCArb-MCNet	**29.3443**	**0.8523**	**0.8932**	**8.8833**	**5.7498**	**5.8167**
	SSPSR	29.6054	0.8629	0.8996	8.6092	5.4015	5.6547
	SSPSR-b	29.4101	0.8540	0.8948	8.8093	5.7598	5.7799
	MCArb-SSPSR	**29.8178**	**0.8698**	**0.9040**	**8.3923**	**5.2883**	**5.5305**
	ERCSR	29.2404	0.8480	0.8908	8.9961	**5.8260**	5.8787
	MCArb-ERCSR	**29.3236**	**0.8515**	**0.8929**	**8.9006**	5.8479	**5.8306**
8×	Bilinear	24.6826	0.4596	0.6704	15.4721	7.8345	4.9211
	Bicubic	25.0540	0.5089	0.6926	14.8011	7.6084	4.7148
	Wavelet	24.4405	0.4609	0.6259	15.9179	8.0325	5.0605
	MCNet	25.4657	0.5802	0.7205	14.0854	7.5622	4.4968
	MCArb-MCNet	**25.5725**	**0.5921**	**0.7279**	**13.9060**	**7.5123**	**4.4429**
	SSPSR	25.6946	0.6094	0.7379	13.7041	**7.1374**	4.3822
	SSPSR-b	25.5789	0.5951	0.7295	13.8895	7.4533	4.4403
	MCArb-SSPSR	**25.7574**	**0.6164**	**0.7417**	**13.5997**	7.1956	**4.3493**
	ERCSR	25.4899	0.5794	0.7222	14.0477	**7.4760**	4.4840
	MCArb-ERCSR	**25.5521**	**0.5915**	**0.7266**	**13.9399**	7.5518	**4.4532**

the quality assessment of hyperspectral image fusion tasks. The optimal values for these indices are $+\infty$, 1, 1, 0, 0, and 0.

C. Experimental Results on Pavia Centre Dataset

The Pavia Centre dataset was acquired by the ROSIS sensor during a flight campaign over Pavia, Northern Italy, in 2001. It encompasses 102 spectral channels ranging from 430 to 860 nm and consists of a 1096 × 1096 pixels image with a geometric resolution of 1.3 m. However, some samples in the images contain no information and must be discarded prior to analysis. We crop the effective region of 1906 × 715 × 102 from the original scene. The left region of the image is selected as testing data, which is cropped into eight 128 × 128 × 102 images. The remaining area of the image is used to form patches of size 64 × 64 × 102 with a 50% overlap for training, with 10% of the training data randomly selected as a validation set. For the three task branches, these patches are utilized to generate LR images through bicubic downsampling at the corresponding scale factors. The sizes of the LR images are 32 × 32 × 102, 16 × 16 × 102, and 8 × 8 × 102.

As illustrated in Table I, we compare the MCArb methods with baseline methods on the Pavia Centre test set. The average values of six quantitative metrics are utilized to evaluate the performance of these methods at scale factors of 4× and 8×. It is observed that MCArb methods, employing a single network for SR at different scale factors, exhibit superior performance compared to baseline methods trained separately for each scale factor. Specifically, at a 4× scale factor, MCArb-MCNet and MCArb-SSPSR significantly outperform their corresponding MCNet [23] and SSPSR [22] across all metrics. Notably, MCArb-SSPSR not only markedly surpasses the directly comparable SSPSR-b but also exceeds the original SSPSR with progressive upsampling. MCArb-ERCSR shows enhanced performance over the ERCSR [27] as baseline in all metrics except for SAM, where it is slightly inferior. At 8×, MCArb-MCNet continues to demonstrate better performance in all metrics. MCArb-SSPSR also outperforms SSPSR-b as baseline in all metrics, with only a slight deficit in SAM compared to SSPSR [22]. The performance comparison of MCArb-ERCSR with the baseline ERCSR [27] at 8× mirrors that at 4×. Overall, MCArb-SSPSR stands out as the best performing among all MCArb methods, showing the greatest improvement over the baseline method in this dataset. Although the other two baseline methods employ 3-D convolution for end-to-end spectral and spatial information extraction, particularly in the reconstruction of spectral information, they do not perform as well as the group-based SSPSR [22] on the Pavia Centre dataset, which is characterized by a multitude of channels and a limited number of samples. The first row of Fig. 9 presents the qualitative outcomes of hyperspectral image reconstruction from the Pavia Centre test set at 4× magnification. For a more effective visual representation, bands 64th, 27th, and 0th of the hyperspectral images are employed as the RGB channels. Our MCArb methods evidently excel in capturing both the low- and high-frequency details when compared to their baseline methods. The second row visualizes the corresponding error maps (absolute difference) with respect to the ground truth. In the error maps (visualized using the Jet colormap), the color distribution indicates the magnitude of reconstruction errors across the spatial domain. Specifically, blue represents lower error values, while red indicates higher errors. The intermediate colors, such as green and yellow, correspond to moderate error levels. The red rectangles highlight the area of interest to compare reconstruction quality in finer detail. In the error maps, we can more clearly observe that our MCArb method demonstrates better SR performance compared to the corresponding baseline methods. Fig. 10 depicts the mean spectral difference for the images displayed in Fig. 9, offering a comprehensive presentation of the spectral reconstruction accuracy. The MCArb methods demonstrate superior performance compared to their baselines. MCArb-SSPSR still achieves the best performance in terms of mean spectral difference, consistent with the SAM data presented in Table I. This further corroborates our analysis presented in Table I and Fig. 9. We found that among the three selected traditional nondeep learning techniques, the bicubic method achieved significantly better results compared to the other two. Therefore, when presenting the outcomes in the figures, only the bicubic method was chosen for comparative display among the traditional techniques.

D. Experimental Results on Chikusei Dataset

The Chikusei dataset was captured by the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Ibaraki, Japan, on July 29, 2014. It encompasses 128 channels within the spectral range of 363–1018 nm and comprises 2517 × 2335 pixels, with a ground sampling distance of 2.5 m. Following prior works [20], [22], we remove invalid information from the edge regions and crop

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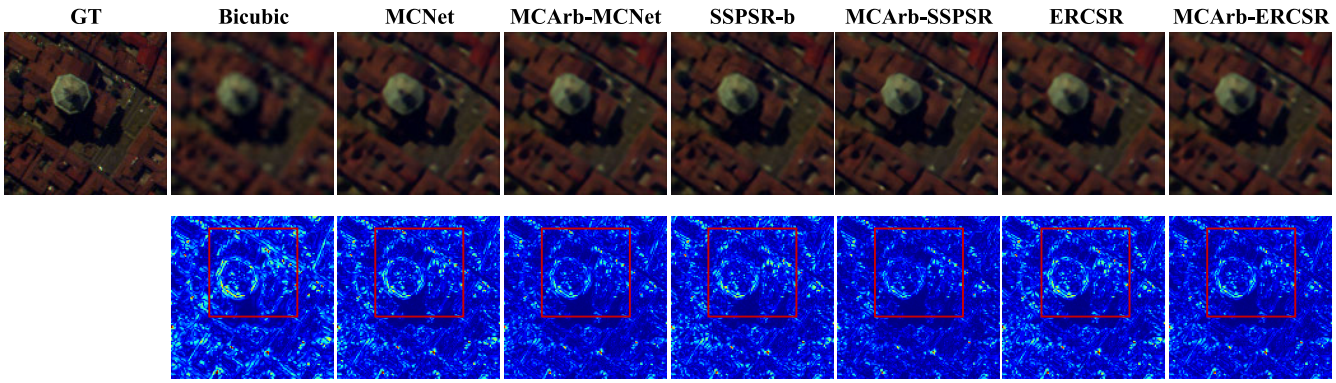


Fig. 9. Reconstructed test hyperspectral image in the Pavia Centre dataset with spectral bands 67–24–0 as R–G–B at scale factor 4×. From left to right, ground truth, results of bicubic, MCNet [23], MCArb-MCNet, SSPSR [22], MCArb-SSPSR, ERCSR [27], and MCArb-ERCSR. The first row shows the reconstructed images from various methods, while the second row visualizes the corresponding error maps (absolute difference) with respect to the ground truth. The red rectangles highlight the area of interest to compare reconstruction quality in finer detail. The ground truth is extracted from a detail-rich region of the Pavia Centre dataset, which includes a variety of buildings, streets, vehicles, and so on. As we can observe, the visual performance of deep learning-based methods is significantly superior to the bicubic method, which performs the best among traditional nondeep learning techniques in terms of metrics. We compared each MCArb method with its corresponding baseline and found that the MCArb methods achieve better detail recovery at the edges of buildings and streets. Overall, MCArb-SSPSR achieves the best visual performance, consistent with its metric performance as shown in Table I.

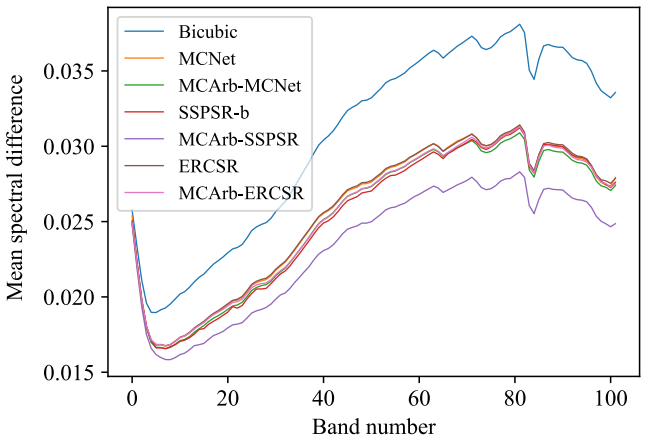


Fig. 10. Mean spectral difference curve of the test hyperspectral image in the Pavia Centre dataset at the scale factors 4×.

out an effective area of 2304 × 2048 × 128 from the original scene. Specifically, the top region of the image is selected as the test data, cropped into 16 nonoverlapping hyperspectral images, each of size 256 × 256 × 128. In the remaining area, we crop overlapping patches of size 64 × 64 × 128 with a 50% overlap for training (10% of the training data are randomly selected as a validation set). For the three task branches, these patches are utilized to generate LR images through bicubic downsampling at the corresponding scale factors. The sizes of those LR images are 32 × 32 × 128, 16 × 16 × 128, and 8 × 8 × 128.

In Table II, we present quantitative results of the MCArb methods and baseline methods on the Chikusei test set. Performance is measured using the averages of six quantitative metrics at scale factors 4× and 8×. MCArb methods, utilizing a single network, continue to outperform the baseline methods trained separately at different scale factors. Specifically, at 4×, MCArb-MCNet significantly surpasses their corresponding MCNet [23] across all metrics. A similar trend is observed for

TABLE II

QUANTITATIVE PERFORMANCE OVER 16 TEST IMAGES WITH DIFFERENT SCALE FACTORS ON THE CHIKUSEI DATASET. BOLD REPRESENTS THE BEST RESULTS

Scale	Method	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	Bilinear	36.7698	0.8921	0.9124	4.0420	3.6432	7.4395
	Bicubic	37.5035	0.9098	0.9244	3.7137	3.4048	6.8393
	Wavelet	36.4090	0.8907	0.9018	4.2173	3.8212	7.7505
	MCNet	39.2166	0.9410	0.9486	3.0627	2.7697	5.5750
	MCArb-MCNet	**39.5406**	**0.9458**	**0.9521**	**2.9569**	**2.7326**	**5.3119**
	SSPSR	39.9136	**0.9520**	0.9560	2.8134	**2.3981**	**5.1699**
	SSPSR-b	39.5761	0.9479	0.9525	2.9216	2.5081	5.3900
	MCArb-SSPSR	**39.9376**	0.9516	**0.9563**	**2.7970**	2.4345	5.1838
	ERCSR	39.2304	0.9409	0.9488	3.0525	2.7406	5.5739
	MCArb-ERCSR	**39.5554**	**0.9458**	**0.9523**	**2.9487**	**2.7368**	**5.3162**
8×	Bilinear	33.9541	0.7976	0.8245	5.6048	5.2956	5.1346
	Bicubic	34.3394	0.8105	0.8377	5.3576	5.0221	4.9184
	Wavelet	33.5336	0.7906	0.8007	5.8856	5.4944	5.3855
	MCNet	35.1240	0.8441	0.8660	4.9032	4.5424	4.4748
	MCArb-MCNet	**35.3545**	**0.8540**	**0.8726**	**4.7986**	**4.4450**	**4.2950**
	SSPSR	35.5664	0.8650	0.8798	4.6373	4.0497	4.2727
	SSPSR-b	35.3343	0.8570	0.8726	4.761	4.2336	4.3973
	MCArb-SSPSR	**35.6480**	**0.8671**	**0.8817**	**4.6033**	**4.0327**	**4.2237**
	ERCSR	35.1294	0.8444	0.8663	4.9010	4.5194	4.4677
	MCArb-ERCSR	**35.3493**	**0.8541**	**0.8725**	**4.7925**	**4.4729**	**4.3154**

MCArb-ERCSR. MCArb-SSPSR notably exceeds SSPSR-b, which employs the same FEN as MCArb-SSPSR but uses subpixel convolution for upsampling. Compared to the original SSPSR [22] with progressive upsampling, our MCArb-SSPSR achieves better PSNR, CC, and RMSE, although slightly lower in SSIM, SAM, and ERGAS. At 8×, all MCArb methods lead over baselines in all six metrics. Overall, our MCArb framework adapts well to FENs based on different strategies and convolution types and enhances baseline models with various upsampling modes. Similar to observations on the Pavia Centre dataset, MCArb-SSPSR performs best on the Chikusei dataset, especially noted for the lowest SAM,

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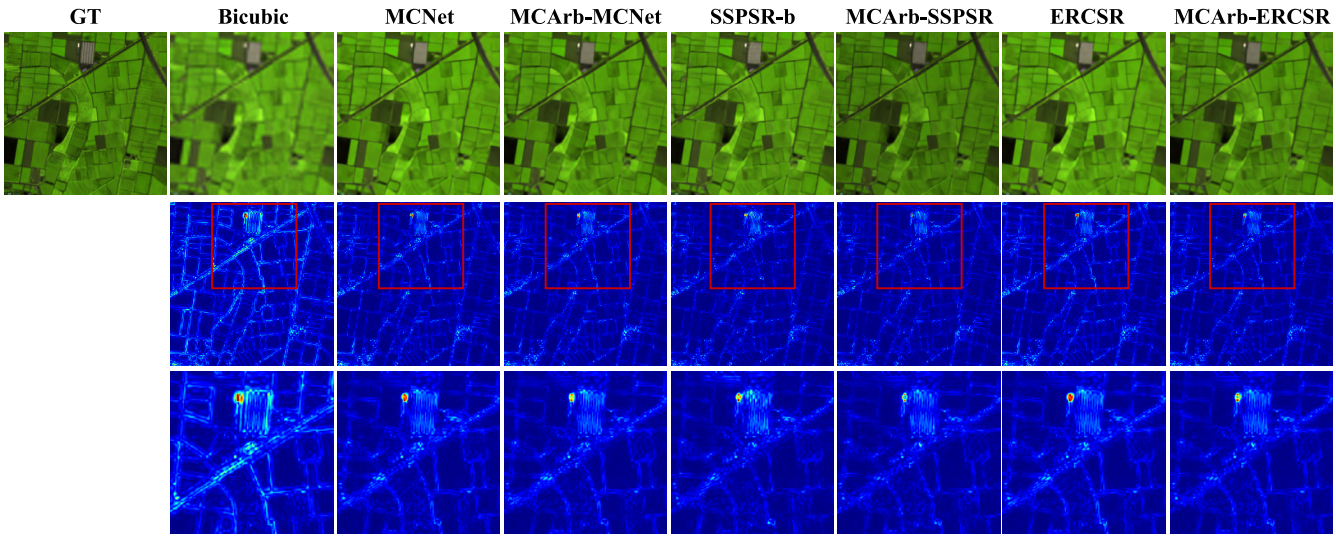


Fig. 11. Reconstructed test hyperspectral image in the Chikusei dataset with spectral bands 70–100–36 as R–G–B at scale factor 4×. The first row shows the reconstructed images from various methods, while the second row visualizes the corresponding error maps (absolute difference) with respect to the ground truth. The red rectangles highlight the area of interest to compare reconstruction quality in finer detail. The third row provides a magnified view of the content within the red rectangles for more detailed analysis. From left to right, ground truth, results of bicubic, MCNet [23], MCArb-MCNet, SSPSR [22], MCArb-SSPSR, ERCSR [27], and MCArb-ERCSR.

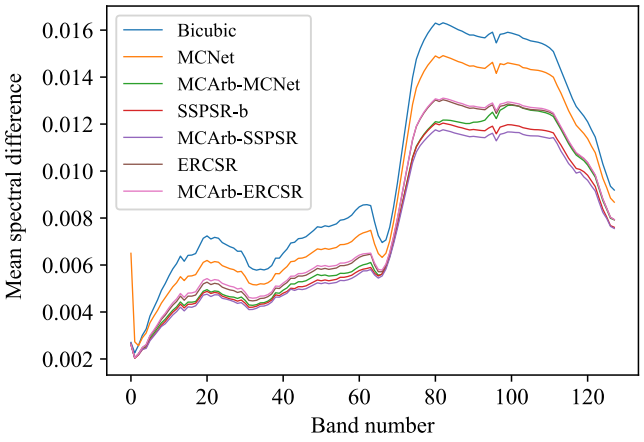


Fig. 12. Mean spectral difference curve of the test hyperspectral image in the Chikusei dataset at the scale factors 4×.

proving its efficacy in datasets with a higher number of channels where group-based networks better extract spectral and spatial information. The first row of Fig. 11 visualizes qualitative results of the reconstructed hyperspectral image from the Chikusei test set at 4×. We select bands 70th, 100th, and 36th as RGB channels for enhanced visualization. The second row visualizes the corresponding error maps (absolute difference) visualized using the Jet colormap with respect to the ground truth. The red rectangles highlight the area of interest to compare reconstruction quality in finer detail. Our MCArb methods demonstrate superior performance in detail, particularly in reconstructing boundaries, as observed in the enlarged red box in the third row. Fig. 12 shows the mean spectral difference for the images displayed in Fig. 11, comprehensively presenting the spectral reconstruction accuracy and corroborating our analysis from Table II and Fig. 11. Our MCArb methods still show superior performance compared to their baseline methods.

E. Experimental Results on Houston 2018 Dataset

The Houston 2018 dataset was acquired using the ITRES CASI 1500 spectral imager over the University of Houston campus and its adjacent urban areas in Houston, TX, USA. It contains 48 channels within the spectral range of 380–1050 nm and consists of 4172 × 1202 pixels, with a ground sampling distance of 1 m. Mirroring the approach taken with the above datasets, we select the top region as test data, cropping 32 nonoverlapping images of size 128 × 128 × 48 for testing. The remaining region of the image is used to form overlapping patches for training, with the patch size being the same as set for the Chikusei datasets (10% of the training data are randomly selected as a validation set). The LR hyperspectral images are generated by bicubic downsampling from these patches for the different scale factors. The sizes of those LR images are 32 × 32 × 48, 16 × 16 × 48, and 8 × 8 × 48.

In Table III, we present the quantitative results of the MCArb methods and baseline methods on the Houston 2018 test set. The performance of these methods is measured using averages of six quantitative metrics. The analysis applicable to the aforementioned dataset also holds true for the performance of MCArb methods on the Houston 2018 dataset. Specifically, regardless of the scale factor at 4× or 8×, MCArb-MCNet and MCArb-ERCSR significantly outperform their respective baseline models across all six metrics. At 8×, MCArb-SSPSR surpasses both the original SSPSR and SSPSR-b, while at 4×, MCArb-SSPSR is slightly inferior to SSPSR and SSPSR-b in SAM, yet leads in the other five metrics. Overall, contrary to our observations on the Pavia Centre and Chikusei datasets, MCArb-MCNet achieves

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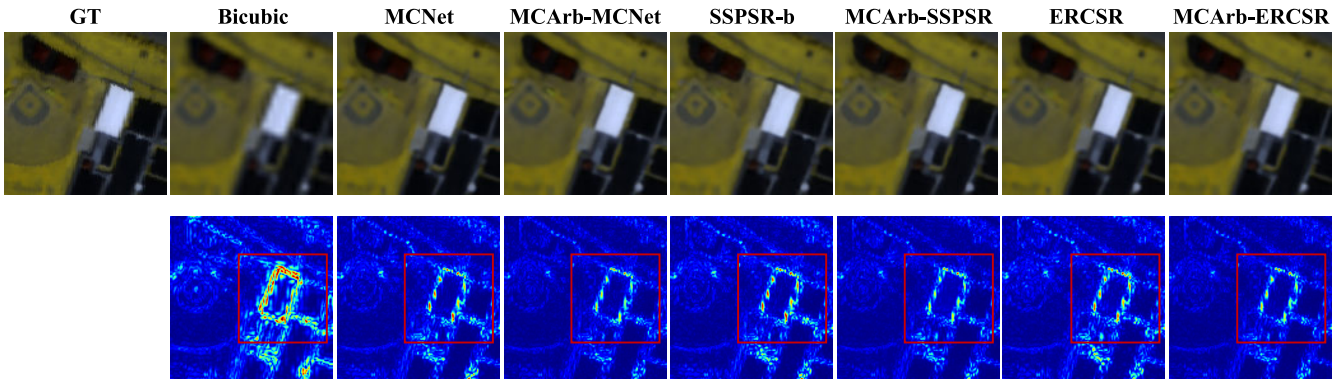


Fig. 13. Reconstructed test hyperspectral image in the Houston 2018 dataset with spectral bands 29–26–19 as R-G-B at scale factor 4×. The first row shows the reconstructed images from various methods, while the second row visualizes the corresponding error maps (absolute difference) with respect to the ground truth. The red rectangles highlight the area of interest to compare reconstruction quality in finer detail. From left to right, ground truth, results of bicubic, MCNet [23], MCArb-MCNet, SSPSR [22], MCArb-SSPSR, ERCSR [27], and MCArb-ERCSR.

TABLE III
QUANTITATIVE PERFORMANCE OVER 32 TEST IMAGES WITH DIFFERENT SCALE FACTORS ON THE HOUSTON 2018 DATASET. BOLD REPRESENTS THE BEST RESULTS

Scale	Method	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	Bilinear	42.7931	0.9582	0.9566	2.0972	2.5152	2.7815
	Bicubic	44.2599	0.9704	0.9669	1.7696	2.1548	2.3491
	Wavelet	42.0007	0.9571	0.9443	2.2849	2.7279	3.0380
	MCNet	46.6818	0.9830	0.9803	1.3370	1.8428	1.7872
	MCArb-MCNet	**47.3451**	**0.9855**	**0.9832**	**1.2376**	**1.6784**	**1.6487**
	SSPSR	**47.1047**	**0.9850**	**0.9824**	**1.2683**	**1.5703**	**1.6872**
	SSPSR-b	46.8401	0.9840	0.9814	1.3070	1.6169	1.7390
	MCArb-SSPSR	47.0361	0.9849	0.9822	1.2740	1.6213	1.6966
	ERCSR	46.5942	0.9825	0.9799	1.3546	1.8259	1.8067
	MCArb-ERCSR	**47.3050**	**0.9854**	**0.9830**	**1.2420**	**1.7225**	**1.6563**
8×	Bilinear	38.1981	0.8847	0.8679	3.5604	4.5125	2.3625
	Bicubic	38.9501	0.8987	0.8842	3.2692	4.0897	2.1692
	Wavelet	37.5183	0.8797	0.8369	3.8394	4.8426	2.5486
	MCNet	40.0855	0.9201	0.9079	2.8704	3.7349	1.9079
	MCArb-MCNet	**40.8440**	**0.9327**	**0.9220**	**2.6318**	**3.4566**	**1.7534**
	SSPSR	40.4438	0.9290	0.9170	2.7407	3.2040	1.8219
	SSPSR-b	40.3160	0.9262	0.9147	2.7767	3.3111	1.8460
	MCArb-SSPSR	**40.5801**	**0.9308**	**0.9193**	**2.6877**	**3.1830**	**1.7895**
	ERCSR	40.0963	0.9201	0.9080	2.8660	3.6802	1.9055
	MCArb-ERCSR	**40.6756**	**0.9301**	**0.9195**	**2.6841**	**3.5346**	**1.7858**

the best results among the three, while MCArb-SSPSR shows the least improvement compared to the baseline. This indicates that in datasets with fewer channels and more samples, an end-to-end strategy with 3-D convolution excels in extracting spectral and spatial information. As illustrated in the first row of Fig. 13, we also visualize the qualitative results of the reconstructed hyperspectral image from the Houston 2018 test set at 4×. Specifically, we select bands 29th, 26th, and 19th as the RGB channels for enhanced visualization. The second row visualizes the corresponding error maps (absolute difference) visualized using the Jet colormap with respect to the ground truth. The red rectangles highlight the area of interest to compare reconstruction quality in finer detail. In the error maps, we can more clearly observe that our MCArb method demonstrates better SR performance compared to the corresponding baseline methods. The advantages of our method are further highlighted in the red box. Fig. 14 displays the mean spectral difference for the images shown in Fig. 13, comprehensively presenting the accuracy of spectral reconstruction quality and corroborating our analysis from Table III and Fig. 13. MCArb methods still show better performance than their baseline methods.

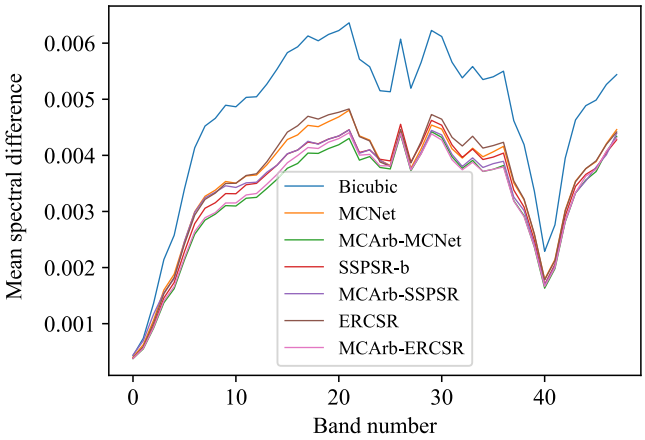


Fig. 14. Mean spectral difference curve of the test hyperspectral image in the Houston 2018 dataset at the scale factors 4×.

F. Experimental Results on Real-World Datasets: CAVE and Harvard

The CAVE dataset, developed by the researchers at Columbia University's Computer Science Department, consists of MSIs captured across 32 scenes. Each scene is imaged in the spectral range from 400 to 700 nm at 10-nm intervals, resulting in 31 spectral channels. Due to spatial–spectral misalignment caused by breathing motion, we excluded two images featuring real people. From the remaining images, we selected 10 to form the test set, dividing these images into nonoverlapping patches of size 256 × 256 × 31. The remaining 20 images were divided into patches of size 64 × 64 × 31 for training, with a 50% overlap (10% of the training data is randomly selected as a validation set). The LR hyperspectral images are generated by bicubic downsampling from these patches for

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TABLE IV
QUANTITATIVE PERFORMANCE OVER TEN TEST IMAGES WITH DIFFERENT SCALE FACTORS ON THE CAVE DATASET. BOLD REPRESENTS THE BEST RESULTS

Scale	Method	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	Bilinear	34.5345	0.9310	0.9745	5.6837	5.2791	6.7545
	Bicubic	35.4925	0.9408	0.9781	5.1365	5.4836	6.1190
	Wavelet	33.5191	0.9296	0.9701	6.2197	5.7649	7.4175
	MCNet	39.0281	0.9632	0.9860	3.7161	4.5679	4.3014
	MCArb-MCNet	**39.3878**	**0.9656**	**0.9866**	**3.5654**	**4.5586**	**4.1152**
	SSPSR	38.7901	**0.9627**	**0.9859**	**3.7669**	**4.6163**	**4.3690**
	SSPSR-b	38.4478	0.9606	0.9849	3.9075	4.7583	4.5317
	MCArb-SSPSR	**38.8181**	0.9620	0.9852	3.7711	4.7262	4.3775
	ERCSR	39.1941	**0.9639**	0.9863	3.6644	**4.4749**	4.2183
	MCArb-ERCSR	**39.2813**	0.9633	**0.9864**	**3.6216**	4.7007	**4.1959**
8×	Bilinear	30.5660	0.8634	0.9458	8.6865	6.9786	5.1096
	Bicubic	31.2517	0.8723	0.9514	8.0720	7.3032	4.7549
	Wavelet	29.5687	0.8548	0.9353	9.5592	7.8052	5.6530
	MCNet	34.1230	**0.9100**	0.9641	6.4147	**6.2641**	3.6082
	MCArb-MCNet	**34.4028**	0.9099	**0.9646**	**6.2874**	6.3958	**3.5162**
	SSPSR	33.7301	0.9034	0.9617	6.6406	7.0703	3.7501
	SSPSR-b	33.7256	0.9015	**0.9620**	6.6453	6.6934	3.7579
	MCArb-SSPSR	**34.1584**	**0.9047**	0.9609	**6.5216**	**6.2761**	**3.6680**
	ERCSR	**34.3315**	**0.9098**	**0.9643**	**6.3363**	**6.1597**	**3.5411**
	MCArb-ERCSR	34.0421	0.9054	0.9640	6.4312	6.8637	3.6215

TABLE V
QUANTITATIVE PERFORMANCE OVER 20 TEST IMAGES WITH DIFFERENT SCALE FACTORS ON THE HARVARD DATASET. BOLD REPRESENTS THE BEST RESULTS

Scale	Method	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	Bilinear	42.0655	0.9297	0.9238	2.9689	3.1626	4.8622
	Bicubic	42.8247	0.9379	0.9307	2.7362	3.1513	4.4850
	Wavelet	41.4303	0.9290	0.9163	3.1818	3.2464	5.1686
	MCNet	44.4191	0.9520	0.9456	2.2887	3.0304	3.7357
	MCArb-MCNet	**44.4975**	**0.9536**	**0.9469**	**2.2591**	**3.0292**	**3.6976**
	SSPSR	44.2865	0.9519	0.9448	2.2964	**3.0143**	3.7743
	SSPSR-b	44.1835	0.9510	0.9437	2.3268	3.0431	3.8153
	MCArb-SSPSR	**44.3727**	**0.9537**	**0.9461**	**2.2706**	3.0511	**3.7287**
	ERCSR	44.4787	0.9532	0.9467	2.2696	**3.0221**	3.7091
	MCArb-ERCSR	**44.5684**	**0.9542**	**0.9471**	**2.2287**	3.0636	**3.6612**
8×	Bilinear	39.0866	0.8810	0.8665	4.1783	3.5771	3.4110
	Bicubic	39.5490	0.8872	0.8747	3.9617	3.6311	3.2385
	Wavelet	38.3353	0.8766	0.8493	4.5663	3.7475	3.6921
	MCNet	40.8937	0.9052	0.8978	3.4127	**3.4998**	2.7861
	MCArb-MCNet	**41.0268**	**0.9071**	**0.9017**	**3.3781**	3.5197	**2.7464**
	SSPSR	40.7653	0.9043	0.8967	3.4526	3.4852	2.8191
	SSPSR-b	40.6581	0.9027	0.8934	3.4846	3.5792	2.8526
	MCArb-SSPSR	**40.9351**	**0.9063**	**0.8994**	**3.4105**	**3.4805**	**2.7842**
	ERCSR	40.9030	0.9049	0.8982	3.4166	**3.4582**	2.7821
	MCArb-ERCSR	**41.0272**	**0.9080**	**0.9018**	**3.3531**	3.6112	**2.7339**

the different scale factors. The sizes of those LR images are $32 \times 32 \times 48$, $16 \times 16 \times 48$, and $8 \times 8 \times 48$.

Similarly, the Harvard dataset, created by researchers at Harvard University, comprises hyperspectral images covering a range of 420–720 nm with a 10-nm spectral interval, producing 31 spectral channels. The images in this dataset span a variety of natural and man-made scenes, both indoor and outdoor. We selected the outdoor portion, consisting of 50 images captured in daylight illumination. Among these, 20 images are used for testing, while the remaining 30 images are used for training. The method of constructing the test and training sets on the Harvard dataset, as well as the corresponding patch sizes, are identical to those set on the CAVE dataset (still 10% of the training data are randomly selected as a validation set). It is noteworthy that each image in the Harvard dataset includes a mask map, where a value of 0 implies that the pixel is masked out. When constructing the test and training sets, we exclude samples containing masked regions. In addition, we remove patches with excessively low mean values, which typically originate from the black parts of the images, to prevent interference with the model's training and testing.

In Tables IV and V, we present the quantitative results of the MCArb methods and baseline methods on the CAVE and Harvard dataset, respectively. The performance of these methods is measured using averages of six quantitative metrics. The main trends in analysis applicable to the aforementioned datasets also hold true for the performance of MCArb methods on these two datasets. Overall, our framework can adapt well when the data source becomes a real-world dataset rather than remote sensing hyperspectral images. As we can see, MCArb-MCNet and MCArb-ERCSR show better performance than MCArb-SSPSR. As analyzed with the Houston2018 dataset, we can conclude that in datasets with fewer channels and more samples, an end-to-end strategy with 3-D convolution performs better in extracting spectral and spatial information than group-based methods.

Notably, on the CAVE dataset, we observe that MCArb-ESRCSR shows less performance than its baseline at 8×, which is the first instance of this occurrence in all our experiments. We believe that unlike remote sensing hyperspectral images and outdoor scenes (Harvard dataset), the content in images of man-made scenes on the CAVE dataset is more concentrated. The input patch for the 8× task branch is comparatively too small, which is a disadvantage for the ERCSR network that emphasizes global information perception. This results in the weakening of the auxiliary task branch's role during the training of MCArb-ESRCSR, leading to the unique case where the 8× performance is less than the baseline on this particular dataset.

G. Ablation Study

In this section, we demonstrate the effectiveness of various strategies and modules proposed in this article. We modify our model and compare different variants. These experiments are conducted on the Chikusei dataset.

1) Effectiveness of Collaborative Strategy: Our framework allows SHSR models to perform SR at arbitrary scales, implying the use of shared-parameter FENs to train multiple SR tasks simultaneously, a feature unachievable in the original model. We aim to evaluate the improvement in SR performance through multitask branches of collaborative training. For this, the training process is kept constant, contrasting single-branch training separately against three-task branch collaborative training. As demonstrated in Table VI,

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TABLE VI

QUANTITATIVE COMPARISONS OF COLLABORATIVE STRATEGY OVER THE TESTING SET ON THE CHIKUSEI DATASET AT THE SCALE FACTORS 4× AND 8×. BOLD REPRESENTS THE BEST RESULTS

Scale	BM.	Variant	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	MCNet	Ours	**39.5406**	**0.9458**	**0.9521**	**2.9569**	**2.7326**	**5.3119**
		w/o CS	39.2120	0.9410	0.9486	3.0622	2.7967	5.5809
	SSPSR	Ours	**39.9376**	**0.9516**	**0.9563**	**2.7970**	**2.4345**	**5.1838**
		w/o CS	39.7469	0.9498	0.9544	2.8599	2.4812	5.2953
	ERCSR	Ours	**39.5554**	**0.9458**	**0.9523**	**2.9487**	**2.7368**	**5.3162**
		w/o CS	39.1718	0.9405	0.9482	3.0750	2.7882	5.6118
8×	Mcnet	Ours	**35.3545**	**0.8540**	**0.8726**	**4.7986**	**4.4450**	**4.2950**
		w/o CS	35.0588	0.8417	0.8638	4.9430	4.5798	4.4992
	SSPSR	Ours	**35.6480**	**0.8671**	**0.8817**	**4.6033**	**4.0327**	**4.2237**
		w/o CS	35.3869	0.8611	0.8743	4.7373	4.2008	4.3572
	ERCSR	Ours	**35.3493**	**0.8541**	**0.8725**	**4.7925**	**4.4729**	**4.3154**
		w/o CS	34.2058	0.8092	0.8318	5.4311	5.1230	5.0020

TABLE VII

QUANTITATIVE COMPARISONS OF 3-D STRUCTURE IN 3DMETAUM OVER THE TESTING SET ON CHIKUSEI DATASET AT THE SCALE FACTORS 4× AND 8×. BOLD REPRESENTS THE BEST RESULTS

Scale	BM.	Variant	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	MCNet	Ours	**39.5406**	**0.9458**	**0.9521**	**2.9569**	**2.7326**	**5.3119**
		w/o 3D	37.0532	0.9074	0.9156	3.9081	3.3650	7.2030
	SSPSR	Ours	**39.9376**	**0.9516**	**0.9563**	**2.7970**	**2.4345**	**5.1838**
		w/o 3D	39.5462	0.9479	0.9523	2.9249	2.5119	5.4271
	ERCSR	Ours	**39.5554**	**0.9458**	**0.9523**	**2.9487**	**2.7368**	**5.3162**
		w/o 3D	38.7320	0.9371	0.9424	3.2115	2.9682	5.9750
8×	Mcnet	Ours	**35.3545**	**0.8540**	**0.8726**	**4.7986**	**4.4450**	**4.2950**
		w/o 3D	33.8268	0.7972	0.8159	5.688	5.2508	5.1977
	SSPSR	Ours	**35.6480**	**0.8671**	**0.8817**	**4.6033**	**4.0327**	**4.2237**
		w/o 3D	35.2602	0.8560	0.8704	4.7974	4.3445	4.4487
	ERCSR	Ours	**35.3493**	**0.8541**	**0.8725**	**4.7925**	**4.4729**	**4.3154**
		w/o 3D	34.6941	0.8348	0.8515	5.1333	4.7648	4.7315

- "BM." represent baseline SHSR models.

the network model developed with the collaborative strategy significantly surpasses the models trained individually for each scale factor, at both 4× and 8×. Ours, in comparison to w/o CS, attains higher PSNR, SSIM, and CC, and lower RMSE, SAM, and ERGAS across various baseline methods, including MCNet [23], SSPSR [22], and ERCSR [27].

This enhancement is attributable to two factors. First, the collaborative strategy exposes the FEN to a more diverse and extensive range of sample inputs, where lower scale factor degraded samples enrich texture information, and higher scale factor samples improve the FEN's capability in capturing the overall image structure, thus boosting its robustness. Second, the varied scale factor inputs to our 3DMetaUM enhance its predictive ability for 3-D convolutional kernel weights compared to a single-scale input.

2) Effectiveness of 3-D Structure in 3DMetaUM: Currently, meta-learning applications in natural image SR predominantly involve the use of meta-learning to predict 2-D convolutional kernel weights [35], [37], [38]. These methods have achieved remarkable results in natural image tasks. However, when directly applied to hyperspectral image SR tasks, they tend to underperform. As shown in Table VII, the meta-method, representing a combination of the original network's FEN with a 2-D meta-based upsampling module, performs significantly worse than the proposed 3DMetaUM, occasionally yielding results comparable to bicubic interpolation. For 3-D data cubes with huge spectral channels, 3-D convolution is more effective compared to 2-D. Moreover, when the baseline methods are MCNet [23] and ERCSR [27], which feature a 3-D convolutional structure in the FEN, the 2-D meta-based upsampling module faces difficulties in handling feature maps produced by 3-D convolution. It is noteworthy that the baseline network employing MCNet [23], without a 3-D structure in MetaUM, undergoes the most severe deterioration in its SR capabilities than ours, its performance almost equating to that of the bicubic method. This indicates that MetaUM fails to learn effective kernel weight prediction for the SR network from the given tasks.

TABLE VIII

QUANTITATIVE COMPARISONS OF AUXILIARY TASK BRANCH SETTING OVER THE TESTING SET ON THE CHIKUSEI DATASET AT THE SCALE FACTORS 4× AND 8×. BOLD REPRESENTS THE BEST RESULTS. (a) ALPHA TIMES SETTING. (b) EQUIVALENT SAMPLE VOLUME SETTING

(a)

Scale	Variant	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	Ours	**39.5406**	**0.9458**	**0.9521**	**2.9569**	**2.7326**	**5.3119**
	w/ 1×	39.2186	0.9408	0.9486	3.0664	2.7828	5.5553
8×	Ours	**35.3545**	**0.8540**	**0.8726**	**4.7986**	**4.4450**	**4.2950**
	w/ 1×	35.1789	0.8480	0.8674	4.8866	4.5507	4.4203

(b)

Scale	Sample	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	100%	39.2118	0.9408	0.9486	3.0671	2.7795	5.5717
	39%	**39.5411**	0.9457	**0.9520**	**2.9566**	2.7327	**5.3031**
	25%	39.5406	**0.9458**	0.9521	2.9569	**2.7326**	5.3119
	0%	39.1968	0.9410	0.9484	3.0758	2.7805	5.5666
8×	100%	35.2074	0.8472	0.8685	4.8689	4.5321	4.4101
	39%	35.3542	0.8539	**0.8729**	**4.7982**	**4.4449**	**4.2310**
	25%	**35.3545**	**0.8540**	0.8726	4.7986	4.4450	4.2950
	0%	35.1798	0.8465	0.8676	4.8976	4.5598	4.4000

3) Analysis of Auxiliary Task Branch: Due to the spatial resolution of hyperspectral images, SR tasks at larger scale factors bear more practical significance. In this article, we designate the smaller scale factor, $\alpha\times$ task branch as the auxiliary task branch. We discuss the choice of $\alpha\times$ and how the equivalent sample volume in this task branch impacts the efficiency of our methods with MCNet [23] as the baseline. We compare $\alpha\times = 1\times$ with $2\times$. One of the primary roles of the $\alpha\times$ branch, we believe, is to enrich the texture information received by the FEN. However, as indicated in Table VIII, the $\alpha\times = 1\times$ method, despite having more texture information, underperforms compared to $2\times$. This can be attributed to the $1\times$ setting reaching a boundary effect in terms of texture detail enhancement, which is less conducive for learning in the 3DMetaUM that relies on scale factors as inputs. The interval between $2\times$, $4\times$, and $8\times$ is more favorable for its learning. Simultaneously, the $1\times$ setting alters the task nature

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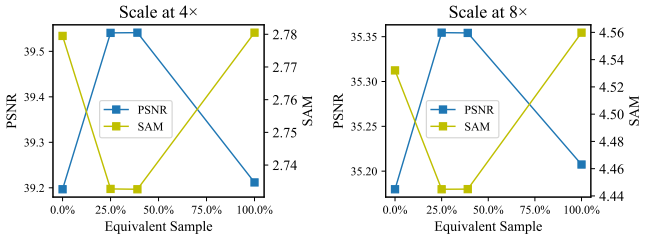


Fig. 15. Results on equivalent sample volume setting.

TABLE IX

QUANTITATIVE COMPARISONS OF 3DSPPM OVER THE TESTING SET ON THE CHIKUSEI DATASET AT THE SCALE FACTORS 4× AND 8×. BOLD REPRESENTS THE BEST RESULTS

Scale	Variant	PSNR↑	SSIM↑	CC↑	RMSE↓	SAM↓	ERGAS↓
4×	Ours	**39.5406**	**0.9458**	**0.9521**	**2.9569**	**2.7326**	**5.3119**
	w/o 3DSPPM	39.3404	0.9425	0.9500	0.9500	2.7232	5.4821
8×	Ours	**35.3545**	**0.8540**	**0.8726**	**4.7986**	**4.4450**	**4.2950**
	w/o 3DSPPM	35.1722	0.8472	0.8670	4.8965	4.5252	4.4209

from SR to sharpening, thereby disrupting the learning of the upsampling module. Considering all factors, a setting of 2× is deemed more appropriate.

At the same time, we explore the impact of varying the equivalent sample volume on the model. The proportion of equivalent samples refers to the use of window cropping on the input samples of the $\alpha\times$ branch during training. For instance, a 16 × 16 window in a 32 × 32 sample equals 25% equivalent samples, and a 20 × 20 window equals 39%. As demonstrated in Table V and Fig. 15, we observe the influence of increasing the equivalent sample volume on the effectiveness of the $\alpha\times$ branch. We believe that without the $\alpha\times$ branch, the FEN would not be able to receive more detailed image inputs. However, when the proportion of equivalent samples in the $\alpha\times$ branch is too high, it will cause 3DMetaUM to be more inclined to learn and adapt to smaller scale factor SR task, resulting in reduced performance of our methods on 4× and 8×. Taking into account the performance, we believe that setting it to 25% is a reasonable choice.

4) Effectiveness of 3DSPPM: Within our proposed framework, we utilize 3DSPPM to minimize the distance between feature maps, thereby facilitating information flow across multiple branches. As illustrated in Table IX, we choose MCNet [23] as the baseline model and compare the performance of our framework with and without the implementation of 3DSPPM. It is observed that excluding 3DSPPM from the training process results in diminished performance of the network.

H. Experimental Results on Feature Shift and Correction

In Fig. 16, we visually demonstrate the feature shift and its correction. It is evident that both SSPSR and MCArb-SSPSR, trained with a fixed scale factor of 4×, extract features from the same 128 × 128 patch of the Chikusei dataset, encompassing the original image and its 2×, 4×, and 8× downsampled versions (as shown in the first column). The third column

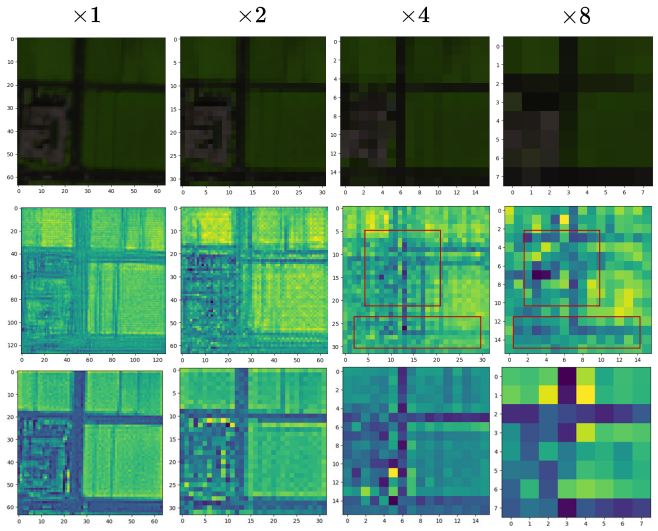


Fig. 16. Results on feature shift and correction.

TABLE X

COMPUTATIONAL PERFORMANCE AND RESOURCE ANALYSIS AT THE INPUT SIZE 1 × 31 × 32 × 32 AND THE SCALE FACTORS 4× AND 8×

scale	method	FLOPs(G)	Params(MB)	Mem(MB)	ExecTime(s)
4×	MCNet	280.2228	8.2935	175.6133	0.0546
	MCArb-MCNet	55.6147	6.6305	458.4209	0.0982
	SSPSR	61.0941	49.1861	148.8755	0.0299
	SSPSR-b	38.8460	49.1861	148.7817	0.0230
	MCArb-SSPSR	38.9225	31.2092	807.5073	0.3047
	ERCSR	278.7565	6.0825	308.7251	0.0456
	MCArb-ERCSR	54.1479	4.4195	493.6284	0.0948
8×	MCNet	2554.6251	11.2935	564.5508	0.1044
	MCArb-MCNet	58.4647	6.6305	1575.3604	0.2808
	SSPSR	195.7627	58.1900	317.2583	0.0618
	SSPSR-b	84.5219	58.1900	316.5396	0.0320
	MCArb-SSPSR	82.4122	31.2092	3179.0093	1.2240
	ERCSR	2606.9900	9.0825	1064.5376	0.0968
	MCArb-ERCSR	110.8276	4.4195	1616.3804	0.2460

distinctly visualizes one channel from the feature maps of MCArb-SSPSR, markedly maintaining consistency in feature extraction across different scale factor downsamples. It is noteworthy that the original SSPSR employs progressive upsampling, resulting in the corresponding feature map dimensions being twice the size of the original image in both length and width.

I. Computational Performance and Resource Analysis

As illustrated in Table X, we can clearly see the computational performance and resource analysis for our MCArb methods and their baselines. This includes floating-point operations (FLOPs) measured in gigaflops, parameters (Params) measured in megabytes, GPU memory usage (Mem) measured in megabytes, and execution time (ExecTime) measured in seconds. Overall, regardless of the scale factor at 4× or 8×, MCArb methods typically show significantly fewer FLOPs and Params but higher Mem and ExecTime.

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Considering that the MCArb methods utilize the baselines' FENs, the major differences lie in the upsampling modules. Compared to the transposed convolution and pixel shuffle in the baselines, which require multiple layers of extensive convolution operations, the 3DMetaUM in MCArb methods only needs to predict the weights of convolutional kernels and perform a few convolution operations. Therefore, MCArb methods have fewer FLOPs and Params. However, meta-learning-based methods require more execution time, and some modules and intermediate variables designed for training in the framework cause higher memory usage during testing.

J. Summary and Discussion of Experimental Results

In summary, our framework can transform various strategy-based deep learning SHSR models into arbitrarily scaled SR models. This enables arbitrarily scaled SR within a single network while enhancing performance. Overall, in most experiments, MCArb methods generally show improved performance compared to the baselines, especially at larger scale factors. This advantage is more evident in spatial evaluation metrics like PSNR. However, when it comes to spectral-related metrics such as SAM, our framework sometimes underperforms.

We believe that there is an inherent tradeoff between spatial and spectral reconstruction quality. Maintaining spectral quality while improving spatial quality is a key goal in SHSR tasks, and our framework achieves this effectively. In addition, compared to the baseline training process, the MCArb methods require consideration of more loss functions, which means that they receive less supervision from spectral information during training, leading to less impressive performance in spectral-related metrics.

As previously analyzed, in terms of dataset applicability, MCArb methods are broadly suitable for remote sensing hyperspectral images and real-world MSIs. However, in specific cases, our framework shows underperformance in some man-made scenes.

V. Conclusion

In this article, the MCArb framework is introduced for arbitrarily scaled hyperspectral image SR using a single network. This framework exploits the potential interdependence among different scale factors by leveraging a meta-learning-based upsampling module and a collaborative strategy. Within MCArb, we initially designed a 3DMetaUM, enabling arbitrarily scaled SR within a single network and realizing the collaborative strategy in our framework. In addition, an SAFAM is integrated after each FEM in the FEN to correct feature shift. We then collaboratively train three SR task branches at different scale factors, enhancing information flow between these branches through a 3DSPPM. The qualitative and quantitative analyses on common hyperspectral image datasets confirm the superiority of our framework. By transforming original state-of-the-art SHSR methods into MCArb methods, we achieve arbitrarily scaled SR with a single network. Moreover, the performance of this single network at different scale factors consistently surpasses that of the models from the original methods, which were trained separately for each factor, especially at higher scale factors. In future endeavors, our work will concentrate on developing flexible methods utilizing meta-learning and collaborative strategies, aimed at harnessing multisource, multimodal, and multi-informational aspects in the field of hyperspectral image processing.

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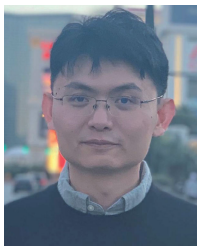
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Mingyang Zhang (Member, IEEE) received the B.S. degree in automation and the Ph.D. degree in pattern recognition and intelligent system from Xidian University, Xi'an, China, in 2012 and 2018, respectively.

Since 2018, he has been a Teacher with Xidian University, where he is currently an Associate Professor. His research interests include computational intelligence and remote sensing image understanding.



Shuang Wu received the B.Eng. degree from Xidian University, Xi'an, China, in 2023, where he is currently pursuing the Ph.D. degree in electronic science and technology with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education.

His research interests include super-resolution and intelligent interpretation of remote sensing images.

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Xiangyu Wang received the B.Eng. degree in intelligence science and technology from Xidian University, Xi'an, China, in 2022, where he is currently pursuing the Ph.D. degree in electronic science and technology with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education.

His research interests include deep learning, computer vision, and intelligent interpretation of remote sensing images.



Maoguo Gong (Fellow, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2003 and 2009, respectively.

Since 2006, he has been a Teacher with Xidian University. He was promoted to an Associate Professor and a Full Professor in 2008 and 2010, respectively, with exceptive admission. He has authored or co-authored over 100 papers in journals and conferences. He holds over 20 granted patents as the first inventor. He is leading or has completed over 20 projects as the principal investigator, funded by the National Natural Science Foundation of China and the National Key Research and Development Program of China. His research interests are broadly in the area of computational intelligence, with applications to optimization, learning, data mining, and image understanding.

Prof. Gong is the Executive Committee Member of Chinese Association for Artificial Intelligence and a Senior Member of Chinese Computer Federation. He was a recipient of the prestigious National Program for Support of the Leading Innovative Talents from the Central Organization Department of China, the Leading Innovative Talent in Science and Technology from the Ministry of Science and Technology of China, the Excellent Young Scientist Foundation from the National Natural Science Foundation of China, the New Century Excellent Talent from the Ministry of Education of China, and the National Natural Science Award of China. He is an Associate Editor or an Editorial Board Member of over five journals, including IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS.



Fenlong Jiang (Member, IEEE) received the B.Eng. and Ph.D. degrees in electronic science and technology from Xidian University, Xi'an, China, in 2017 and 2022, respectively.

Since 2023, he has been a Lecturer with the School of Computer Science and Technology, Xidian University. His research interests include deep learning, computer vision, and remote sensing image interpretation and processing.



Yu Zhou received the B.Eng. and M.S. degrees from Air Force Engineering University, Xi'an, China, in 2005 and 2008, respectively, and the Ph.D. degree from the National University of Defense Technology, Changsha, China, in 2013.

Since 2021, he has been a Teacher with Xidian University, where he is currently a Professor. His research interests include computational intelligence and multiparty collaborative learning.



Yue Wu (Senior Member, IEEE) received the B.Eng. and Ph.D. degrees from Xidian University, Xi'an, China, in 2011 and 2016, respectively.

Since 2016, he has been a Teacher with Xidian University, where he is currently a Professor. He has authored or co-authored more than 90 papers in refereed journals and proceedings. His research interests include computational intelligence and its applications.

Dr. Wu is the Secretary General of Chinese Association for Artificial Intelligence-Youth Branch, the Chair of CCF YOCSEF Xi'an, and a Senior Member of Chinese Computer Federation. He is an Editorial Board Member for over five journals, including *Remote Sensing*, *Applied Sciences*, *Electronics*, and *Mathematics*.

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